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Dialyzer manufacturing tool

Abstract

A dialyzer housing manufacturing system includes a molding device configured to mold a dialyzer housing, and a tool coupled to a robotic arm and configured to retrieve the dialyzer housing from the molding device after the dialyzer housing is molded. The tool includes a frame, a first suction cup connected to a first portion of the frame, and a second suction cup connected to a second portion of the frame, the second suction cup being oriented about 70 degrees to about 110 degrees relative to the first suction cup.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional application of and claims the benefit of priority to U.S. application Ser. No. 16/703,287, filed on Dec. 4, 2019, the contents of which are hereby incorporated by reference.

TECHNICAL FIELD

(1) This disclosure relates to a robotic arm tool for use in dialyzer manufacturing.

BACKGROUND

(2) Hemodialysis is a treatment used to support a patient with insufficient renal function. During hemodialysis, a patient's blood is passed through a dialyzer of a dialysis machine while also passing a dialysis solution or dialysate through the dialyzer. Dialyzers include a housing and a semi-permeable membrane contained within the housing of the dialyzer. The semi-permeable membrane separates the blood from the dialysate within the dialyzer and allows diffusion and osmosis exchanges to take place between the dialysate and the blood stream. The housings of dialyzers are typically manufactured using an injection molding process.

SUMMARY

(3) In one aspect, a dialyzer housing manufacturing system includes a molding device configured to mold a dialyzer housing, and a tool coupled to a robotic arm and configured to retrieve the dialyzer housing from the molding device after the dialyzer housing is molded. The tool includes a frame, a first suction cup connected to a first portion of the frame, and a second suction cup connected to a second portion of the frame, the second suction cup being oriented about 70 degrees to about 110 degrees relative to the first suction cup.

(4) Embodiments can include one or more of the following features in any combination.

(5) In certain embodiments, the first and second suction cups are fluidly coupled to a vacuum source.

(6) In some embodiments, the dialyzer housing manufacturing system further includes a third suction cup connected to the first portion of the frame, a fourth suction cup connected to the first portion of the frame, a fifth suction cup connected to the first portion of the frame, a sixth suction cup connected to the second portion of the frame, a seventh suction cup connected to the second portion of the frame; and an eighth suction cup connected to the second portion of the frame, the sixth, seventh, and eighth suction cups being oriented about 70 degrees to about 110 degrees relative to the third, fourth, and fifth suction cups.

(7) In certain embodiments, the molding device is configured to mold two dialyzer housings.

(8) In some embodiments, the tool is configured to simultaneously retrieve two dialyzer housings from the molding device.

(9) In certain embodiments, the molding device is an injection molding device.

(10) In some embodiments, the tool is rotatable between a first position and a second position.

- (11) In certain embodiments, the dialyzer housing manufacturing system further includes a pneumatic cylinder, and a rotation pin, wherein the rotation pin couples the tool to the robotic arm, and the tool is configured to rotate about the rotation pin in response to a force applied to the tool by the pneumatic cylinder.
- (12) In some embodiments, a width of the tool in the first position is about 16 cm to about 17 cm.
- (13) In certain embodiments, the molding device is configured to open a pair of mold halves between about 200 mm to about 240 mm after molding the dialyzer housing.
- (14) In some embodiments, a width of the tool in the second position is about 35 cm to about 36 cm.
- (15) In certain embodiments, the mold includes an alignment pin coupled to a first half of the mold, and the alignment pin remains partially inserted into a second half of the mold when the mold is opened after molding the dialyzer housing.
- (16) In some embodiments, the dialyzer housing manufacturing system further includes a cooling table for cooling the dialyzer housing.
- (17) In certain embodiments, the dialyzer housing manufacturing system further includes a storage container for storing the dialyzer housing.
- (18) In a further aspect, a method includes opening a mold to expose a first dialyzer housing, coupling the first dialyzer housing to a first portion of a tool, moving the tool to remove the first dialyzer housing from the mold, rotating the tool about 70 degrees to about 110 degrees to orient the first portion of the tool in a first direction, placing the first dialyzer housing at a first location using the tool, rotating the tool about 70 degrees to about 110 degrees to orient a second portion of the tool in the first direction, coupling a second dialyzer housing at the first location to the second portion of the tool, and placing the second dialyzer housing at a second location using the tool.
- (19) Embodiments can include one or more of the following features in any combination.
- (20) In some embodiments, the mold is opened about 200 mm to about 240 mm.
- (21) In certain embodiments, coupling the first dialyzer housing to a first portion of a tool includes inserting the tool between a first half of the mold and a second half of the mold.
- (22) In some embodiments, inserting the tool between a first half of the mold and the second half of the mold includes extending a robotic arm coupled to the tool between the first half of the mold and the second half of the mold.
- (23) In certain embodiments, coupling the first dialyzer housing to the first portion of the tool includes positioning one or more suction cups coupled to the first portion of the tool proximate the first dialyzer housing, and applying vacuum suction through an opening in each of the one or more suction cups.
- (24) In some embodiments, placing the first dialyzer housing at a first location using the tool includes positioning the first dialyzer housing proximate the first location using the tool, and stopping the application of vacuum suction through the opening of each of the one or more suction cups.
- (25) In certain embodiments, coupling a second dialyzer housing at the first location to the second portion of the tool includes positioning one or more suction cups coupled to the second portion of the tool proximate the second dialyzer housing, and applying vacuum suction through an opening in each of the one or more suction cups.
- (26) In some embodiments, placing the second dialyzer housing at a second location using the tool includes positioning the second dialyzer housing proximate the second location using the tool, and stopping the application of vacuum suction through the opening of each of the one or more suction cups.
- (27) In certain embodiments, the method further includes coupling a third dialyzer housing to the first portion of the tool, and moving the tool to remove the third dialyzer housing from the mold, wherein the first dialyzer housing and the third dialyzer housing are removed from the mold simultaneously.

- (28) In some embodiments, the method further includes coupling a fourth dialyzer housing at the first location to the second portion of the tool, and placing the fourth dialyzer housing at the second location using the tool, wherein the second dialyzer housing and the fourth dialyzer housing are placed at the second location simultaneously.
- (29) In certain embodiments, the first location comprises a cooling table.
- (30) In some embodiments, the second location comprises a storage container.
- (31) In a further aspect, a device for removing a dialyzer housing from a mold includes a tool coupled to a robotic arm, and a pin rotatably coupling the tool to the robotic arm. The tool includes a frame, a first suction cup connected to a first portion of the frame, and a second suction cup connected to a second portion of the frame, the second suction cup being oriented about 70 degrees to about 110 degrees relative to the first suction cup.
- (32) Embodiments can include one or more of the following features in any combination.
- (33) In certain embodiments, the first and second suction cups are fluidly coupled to a vacuum source.
- (34) In some embodiments, the device further includes a third suction cup connected to the first portion of the frame, a fourth suction cup connected to the first portion of the frame, a fifth suction cup connected to the first portion of the frame, a sixth suction cup connected to the second portion of the frame, a seventh suction cup connected to the second portion of the frame, and an eighth suction cup connected to the second portion of the frame, the sixth, seventh, and eighth suction cups being oriented about 70 degrees to about 110 degrees relative to the third, fourth, and fifth suction cups.
- (35) In certain embodiments, the tool is configured to rotate about 70 degrees to about 110 degrees between a first position and second position about the pin.
- (36) In some embodiments, a width of the tool in the first position is about 16 cm to about 17 cm.
- (37) In certain embodiments, a width of the tool in the second position is about 35 cm to about 36 cm.
- (38) In a further aspect, a dialyzer housing manufacturing system includes a molding device configured to mold a dialyzer housing, and a tool coupled to a robotic arm and configured to retrieve the dialyzer housing from the molding device after the dialyzer housing is molded. The tool includes a frame, a first suction cup connected to a first portion of the frame, and a second suction cup connected to a second portion of the frame, wherein the tool is rotatable between a first position in which the first suction cup extends in a first direction and a second position in which the second suction cup extends in the first direction, and the width of the tool in the second position is greater than the width of the tool in the first position.
- (39) The width of the tool in the first position is measured linearly from the first suction cup to an opposite edge of the tool, and the width of the tool in the second position is measured linearly from the second suction cup to an opposite edge of the tool.
- (40) Advantages of the systems, devices, and methods described herein include reduced wear on the injection molding device. For example, by using a rotatable arm tool to minimize the amount that the mold must be opened during removal of a dialyzer housing from the mold (“de-molding”), the amount of wear on the injection molding device is reduced. In addition, by using a rotatable arm tool to minimize the amount that the mold must be opened during de-molding, the alignment pins of the molding device can remain engaged during de-molding, which reduces the risk of damage to the injection molding device. Another advantage is that the overall time required to perform injection molding of the dialyzer housing is reduced by using a rotatable arm tool to minimize the amount the mold must be opened during de-molding.
- (41) Other aspects, features, and advantages will be apparent from the description and drawings, and from the claims.
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Description

DESCRIPTION OF DRAWINGS

- (1) FIG. 1 depicts a system for manufacturing a dialyzer housing that includes a molding device and a robotic arm tool.
- (2) FIG. 2 is a perspective view of the robotic arm tool of the system of FIG. 1 in a first position.
- (3) FIG. 3 is a perspective view of the robotic arm tool of the system of FIG. 1 in a second position.
- (4) FIG. 4 is a side view of the molding device of the system of FIG. 1.
- (5) FIG. 5 is a front view of a portion of the molding device of the system of FIG. 1.
- (6) FIGS. 6-21 depict an example process for manufacturing a dialyzer housing using the system of FIG. 1.

DETAILED DESCRIPTION

- (7) Referring to FIG. 1, a dialyzer manufacturing system **100** includes an injection molding device **102**, a robotic arm **104**, a cooling table **106**, and a storage container **108**.
- (8) As depicted in FIG. 1, the injection mold includes two mold halves **110**, **112**. As described in further detail herein, the mold halves **110**, **112** can move within the injection molding device **102** to form a cavity in which a dialyzer housing **124** can be molded. For example, as depicted in FIG. 1, during injection molding of the dialyzer housing **124**, the mold halves **110**, **112** are pressed together and molten resin is injected into the cavity formed by the mold halves **110**, **112** to mold the dialyzer housing **124**. Once the dialyzer housing **124** is molded, the mold halves **110**, **112** open to expose the dialyzer housing **124** and allow for retrieval of the dialyzer housing **124** from the injection molding device **102**.
- (9) Once the dialyzer housing has been formed by the injection molding device **102**, the robotic arm **104** is used to retrieve the dialyzer housing **124** from the injection molding device **102**. As depicted in FIG. 1, an arm tool **114** is coupled to an end of the robotic arm **104**. As described in further detail herein, the arm tool **114** can apply a suction force to the dialyzer housings **124** to remove the dialyzer housings **124** from the injection molding device **102**.
- (10) Once the dialyzer housing **124** has been removed from the injection molding device **102** using the arm tool **114**, the robotic arm **104** and arm tool **114** are used to place the dialyzer housing **124** on the cooling table **106**. For example, the arm tool **114** can rotate about the end of the robotic arm **104** to position a dialyzer housing **124** coupled to the arm tool **114** on the cooling table **106**. The cooling table **106** is configured to provide air flow around the dialyzer housing **124** to reduce the temperature of the surface of the newly molded dialyzer housing **124**. As depicted in FIG. 1, the cooling table **106** includes multiple cooling racks **138**, which allows for multiple dialyzer housings **124** to be positioned on the cooling table **106**.
- (11) Once the surface of the dialyzer housing **124** has cooled to a temperature ranging from about 115° C. to about 125° C., a portion of the arm tool **114** couples to the dialyzer housing **124**. The robotic arm **104** and the arm tool **114** lifts the dialyzer housing **124** off the cooling table **106** and places the dialyzer housing **124** within the storage container **108**. Once the storage container **108** is filled with dialyzer housings, a new, empty storage container is provided, and the filled storage container **108** can be used to store or ship the dialyzer housings packed within the storage container **108**.
- (12) Still referring to FIG. 1, the robotic arm **104** of the dialyzer housing manufacturing system **100** includes a base **116**, a lateral boom **118**, and a vertical projection **120**. The base **116** is stationary relative to the injection molding device **102** and includes a set of tracks **122** extending along the length of a top surface of the base **116**. The position of the arm tool **114** can be adjusted by moving various components of the robotic arm **104**. For example, the vertical projection **120** can traverse along the length on the lateral boom **118** and the lateral boom can traverse crosswise along the length of the base **116** by traveling along the tracks **122**. In addition, the vertical projection **120** is

configured to extend to lower the arm tool **114** and retract to raise the arm tool **114**. By coordinating the movements of the lateral boom **118** and the vertical projection **120**, the arm tool **114** can be precisely positioned within three dimensional space.

(13) As depicted in FIG. **1**, the dialyzer manufacturing system **100** also includes a set of controllers **160**, **162**. A first controller **160** is configured to control the injection molding device **102**, and a second controller **162** is configured to control the robotic arm **104** and the arm tool **114**. The controllers **160**, **162** are communicatively coupled with each other and send signal to one another to coordinate the movements of the injection molding device **102**, the robotic arm **104**, and the arm tool **114**. By controlling the timing and movements of the injection molding device **102**, the robotic arm **104**, and the arm tool **114**, the controllers **160**, **162** enable the arm tool **114** to engage and move dialyzer housings **124** throughout the dialyzer housing manufacturing system **100**. For example, the injection molding device controller **160** signals the robotic arm controller **162** when the molds **110**, **112** are in an open position and the robotic arm controller **162** controls the robotic arm **104** and arm tool **114** to retrieve the dialyzer housings from the molds **110**, **112**.

(14) Further, the injection molding device **102** and the robotic arm **104** each include rotary encoder(s) (not shown) that are communicably coupled to the controllers **160**, **162**, and signals received by the controllers **160**, **162** from the rotary encoder(s) can be used to determine the spatial positioning of the components of the injection molding device **102** and robotic arm **104**. For example, rotary encoder(s) are used to measure the number of rotations of the motor(s) of the robotic arm **104** has completed. The controller **162** can determine the direction and distance that the lateral boom **118** and/or vertical projection **120** of the robotic arm **104** has travelled based on the number of rotations that the motor(s) of the robotic arm **104** has completed, as detected by the rotary encoder(s). Based on determining the direction and distance that the lateral boom **118** and/or vertical projection **120** has travelled based on the signals received from the rotary encoder(s), the controller **162** can determine the position of the arm tool **114** in three-dimensional space. Similarly, rotary encoders are used to measure the number of rotations of the motor(s) used to move the mold **110**, **112** has completed. The controller **160** can determine the direction and distance that the mold half **112** of the injection molding device **102** has travelled based on the number of rotations that the motor(s) of the injection molding device **102** has completed, as detected by the rotary encoder(s). Each of the components of the injection molding device **102** and the robotic arm **104** are configured to move predetermined distances throughout the process cycle in order to conduct molding and transporting the dialyzer housings. In some embodiments, the rotary encoders of the injection molding device **102** and the robotic arm **104** each determine the number of rotations that the motor(s) of the injection molding device **102** and the robotic arm **104**, respectively, have completed based on signals received from a proximity switch(es) communicably coupled to the encoders. In some embodiments, the rotary encoders of the injection molding device **102** and the robotic arm **104** each determine the number of rotations that the motor(s) of the injection molding device **102** and the robotic arm **104**, respectively, have completed based on magnets of the rotary encoders.

(15) FIG. **2** depicts a perspective view of the arm tool **114** in a first position **200**. As depicted in FIG. **2**, the arm tool **114** includes eight suction cups **230**, **232**, **234**, **236**, **238**, **240**, **242**, **244** coupled to a frame **280**.

(16) Each suction cup **230**, **232**, **234**, **236**, **238**, **240**, **242**, **244** is configured to couple to a dialyzer housing **124** that is formed by the injection molding device **102**. For example, each suction cup **230**, **232**, **234**, **236**, **238**, **240**, **242**, **244** includes an opening through its center, and each suction cup **230**, **232**, **234**, **236**, **238**, **240**, **242**, **244** is fluidly coupled to a vacuum source, enabling suction to be applied through the center of each suction cup **230**, **232**, **234**, **236**, **238**, **240**, **242**, **244**. As described in further detail herein, a dialyzer housing can be coupled to the suction cups **230**, **232**, **234**, **236**, **238**, **240**, **242**, **244** by applying suction through the center of the respective suction cup.

(17) The suction cups **230**, **232**, **234**, **236**, **238**, **240**, **242**, **244** are divided into a first set of suction

cups **202**, which includes suction cups **230, 232, 234, 236**, and a second set of suction cups **204**, which includes suction cups **238, 240, 242, 244**. As depicted in FIG. 2, the first set of suction cups **202** is coupled to a first portion **282** of the frame **280**, and the second set of suction cups **204** is coupled to a second portion **284** of the frame **280**. As can be seen in FIGS. 2 and 3, the first set of suction cups **202** are coupled to the frame **280** such that the first set **202** is oriented about 90 degrees relative to the second set of suction cups **204**.

(18) Referring to FIG. 1, the system **100** includes a vacuum source **150** that is fluidly coupled to each set of suction cups **202, 204** via a vacuum line **152** that extends along the vertical projection **120**. The controller **160** controls the application of vacuum suction through each set of suction cups **202, 204** to allow for selective engagement of dialyzer housings to the sets of suction cups **202, 204**. The vacuum source **150** applies a vacuum pressure in a range of about 0.35 MPa to about 0.50 MPa to the ends of each of the suction cups. Any of various suitable pumps can be used as the vacuum source **150**, such as a suction pump, a positive displacement pump, a venturi pump, etc.

(19) The vacuum source **150** is communicatively coupled to a controller **160**, which controls the timing of the application of suction by the vacuum source **150** through the sets **202, 204** of suction cups of the arm tool **114**. For example, the controller **160** can coordinate the vacuum suction with the movements of the robotic arm **104** and the arm tool **114** in order to selectively apply vacuum suction through one of the sets of suction cups **202, 204** when the respective set is positioned proximate a dialyzer housing to be moved by the arm tool **114**.

(20) As depicted in FIG. 2, the first portion **282** of the frame **280** forms a rectangular platform **206** and the first set of suction cups **202** is attached to the rectangular platform **206**. When the arm tool **114** is in a first position **200** (as depicted in FIG. 1), the rectangular platform **206** is positioned such that the first set of suction cups **202** coupled to the platform **206** point sideways relative to the vertical projection **120** of the robotic arm **104**. As depicted in FIG. 2, the first set of suction cups **202** includes two pairs of suction cups **208, 210**. Each suction cup **230, 232, 234, 236** in the first set **202** is coupled to the rectangular platform **206** proximate a respective corner of the rectangular platform **206**.

(21) The first set of suction cups **202** is configured to simultaneously couple to two dialyzer housings, with the first pair of suction cups **208** coupling to a first dialyzer housing and the second pair of suction cups **210** coupling to a second dialyzer housing. As previously discussed, each of the suction cups in the first set **202** includes an opening therethrough. The first set of suction cups **202** are fluidly coupled to the vacuum line **152**, which is coupled to the vacuum source **150**, and vacuum suction can be provided through the vacuum line **152** to the first set of suction cups **202** in order to couple a pair of dialyzer housings to the first set of suction cups **202**.

(22) In some examples, the rectangular platform **206** has a total width of about 9 cm to about 11 cm (e.g., about 10.16 cm), a total length of about 16 cm to about 17 cm (e.g., about 16.51 cm), and a total thickness of about 1 cm to about 3 cm (e.g., about 2.54 cm).

(23) Still referring to FIG. 2, the second portion **284** of the frame **280** forms a U-shaped platform **212**, and the second set of suction cups **204** is attached to a U-shaped platform **212**. The U-shaped platform **212** includes a first post **214**, a second post **216**, and a connector bar **218**. The first post **214** is coupled to a first end of the connector bar **218** and the second post **216** is coupled to a second end of the connector bar **218** opposite the first post **214**. As depicted in FIG. 2, the longitudinal axis of the connector bar **218** is substantially perpendicular to the longitudinal axis of each of the posts **214, 216**. When the arm tool **114** is in the first position **200** (as depicted in FIG. 2), the longitudinal axis of each of the posts **214, 216** is substantially parallel to the longitudinal axis of the vertical projection **120** of the robotic arm **104**. As depicted in FIG. 2, the rectangular platform **206** is coupled to the connector bar **218** of the U-shaped platform **212**.

(24) In some examples, each post **214, 216** of the U-shaped platform **212** has a total width of about 7 cm to about 8 cm (e.g., about 7.62 cm), a total length of about 22 cm to about 24 cm (e.g., about 25.4 cm), and a total thickness of about 1 cm to about 3 cm (e.g., about 3.81 cm). In some

examples, the connector bar **218** of the U-shaped platform **212** has a total width of about 7 cm to about 8 cm (e.g., about 7.62 cm), a total length of about 16 cm to about 17 cm (e.g., about 16.51 cm), and a total thickness of about 1 cm to about 3 cm (e.g., about 2.54 cm).

(25) Similar to the first set of suction cups **202**, the second set of suction cups **204** also includes a first pair of suction cups **220** and a second pair of suction cups **222**, totaling four suction cups in the second set **204**. As depicted in FIG. 2, the first pair of suction cups **220** of the second set **204** is coupled to an end of the first post **214**, and the second pair of suction cups **222** of the second set **204** is coupled to an end of the second post **216**.

(26) As depicted in FIG. 2, the second set of suction cups **204** is configured to couple to two dialyzer housings, with the first pair of suction cups **220** coupling to a first dialyzer housing **224** and the second pair of suction cups **222** coupling to a second dialyzer housing **226**. For example, as previously discussed, each of the suction cups in the second set **204** includes an opening therethrough. In addition, the posts **214**, **216** each include a vacuum line **154**, **156**, respectively, coupled to vacuum line **152** to allow suction to be applied through second set of suction cups **204** to couple a pair of dialyzer housings to the second set of suction cups **204**.

(27) FIG. 3 depicts the arm tool **114** in a second position **300**. As shown in FIG. 3, when the arm tool **114** is in the second position **300**, the longitudinal axis of each of the posts **214**, **216** of the U-shaped platform **212** are perpendicular to the longitudinal axis of the vertical projection **120** of the robotic arm **104**. In addition, when the arm tool **114** is in the second position **300**, the first set of suction cups **202** coupled to the rectangular platform **206** are facing downward.

(28) As depicted in FIG. 3, the arm tool **114** is coupled to an end of the vertical projection **120** of the robotic arm **104** using a pin connector **302**, and is rotatable about the end the vertical projection **120** via the pin connector **302**. For example, the arm tool **114** can rotate about 0 degrees to about 90 degrees about the pin connector **302**. In some embodiments, the arm tool **114** rotates about 90 degrees about the pin connector **302** between the first position **200** (depicted in FIG. 2) and the second position **300** (depicted in FIG. 3). The arm tool **114** also includes a pneumatic cylinder (not shown), which applies a force to an end of the arm tool **114** and causes the arm tool to rotate about the pin connector **302**. The controller **160** controls and coordinates the rotation of the arm tool **114** between the first position **200** and the second position **300** during manufacturing and packaging the dialyzer housings. For example, the controller **160** signals the pneumatic cylinder of the arm tool **114** to extend or retract to rotate the arm tool **114** between the first position **200** and the second position **300**.

(29) As can be seen in FIGS. 2 and 3, the width **270** of the profile of the arm tool **114** in the first position **200** is smaller than the width **370** of the profile of the arm tool **114** in the second position **300**. For example the width **270** of the profile of the arm tool **114** in the first position **200** can be about 16 cm to about 17 cm (e.g., about 16.5 cm), and the width **370** of the profile of the arm tool **114** in the second position **300** can be about 35 cm to about 36 cm (e.g., about 35.5 cm). The width **270** of the profile of the arm tool **114** in the first position **200** can be about 18 cm to about 20 cm less than the width **370** of the profile of the arm tool **114** in the second position **300**. As depicted in FIG. 2, the width **270** of the tool **114** in the first position **200** is measured linearly from the first set of suction cups **202** to an opposite edge of the tool. As depicted in FIG. 3, the width **370** of the tool **114** in the second position **300** is measured linearly from the second set of suction cups **204** to an opposite edge of the tool **114**.

(30) Rotating the arm tool **114** between the first position **200** and the second position **300** reduces the amount the injection molding device **102** must be opened to enable the arm tool **114** to fit between the molds **110**, **112**, while still allowing the arm tool **114** to reach the bottom of the storage container **108** to place the dialyzer housings within the storage container **108**. Minimizing the amount the injection molding device **102** must be opened to enable the arm tool **114** to remove the dialyzer housing **124** from the mold halves **110**, **112** can reduce the wear on the molds **110**, **112**. In addition, minimizing the amount the mold halves **110**, **112** must be opened during de-molding can

reduce the risk of misalignment of the mold halves **110**, **112**, and can thus reduce the risk of damage to the injection molding device **102**. Further, minimizing the amount the injection molding device **102** must be opened during de-molding can reduce the time required to open the molding injection device **102** during de-molding, which can reduce the overall time required to manufacture the dialyzer housing **124**.

(31) FIG. **4** depicts a side view of the injection molding device **102** of the dialyzer housing manufacturing system **100**. As depicted in FIG. **4**, the injection molding device **102** includes a first mold half **110** and a second mold half **112**. The mold halves **110**, **112** of the injection molding device **102** are configured to form two dialyzer housings at the same time.

(32) FIG. **5** depicts a front view of the first mold half **110** of the injection molding device **102**. As depicted in FIG. **5**, the first mold half **110** includes two cavities **502**, **504** and multiple alignment pins **510**, **512**, **514**, **516**, **520**, **522**, **524**, **526**.

(33) The cavities **502**, **504** are each used to form a dialyzer housing **124**. The second mold half **112** includes two corresponding cavities (not shown). During injection molding, the cavities **502**, **504** of the first mold half **110** and the cavities of the second mold half **112** are aligned, the mold halves **110**, **112** are positioned against one another, and molten material is injected into the cavities. As the injected material cools, the material takes the form of the cavities in the first and second mold halves **110**, **112** to form the dialyzer housings **124**. The dialyzer housings **124** can be formed of any of various different medical grade materials. Examples of such materials include polycarbonate, polypropylene, etc.

(34) As depicted in FIG. **5**, the first mold half **110** includes a first set of four mold alignment pins **510**, **512**, **514**, **516** and a second set of mold alignment pins **520**, **522**, **524**, **526** projecting outward from the interior surface **518** of the first mold half **110**. The mold alignment pins **510**, **512**, **514**, **516**, **520**, **522**, **524**, **526** ensure proper alignment and a secure fitting between the two mold halves **110**, **112** during injection molding. For example, when the mold halves **110**, **112** of the injection molding device **102** are properly aligned, the mold alignment pins **510**, **512**, **514**, **516**, **520**, **522**, **524**, **526** align with and are inserted into corresponding openings in the second mold half **112** (not shown).

(35) As depicted in FIG. **4**, once the injection molding is complete, the second mold half **112** is moved apart from first mold half **110** by a predetermined distance **450** to expose the dialyzer housings formed by the injection molding device **102**. In some cases, the second mold half **112** is moved about 230 mm apart from the first mold half **110** to accommodate the insertion of the arm tool **114** (positioned in the first position **200**) between the mold halves **110**, **112**. The arm tool **114** is used to remove the dialyzer housings **424** from the second mold half **112**.

(36) As depicted in FIG. **4**, the mold alignment pins **510**, **512**, **514**, **516** and core alignment pins **520**, **522**, **524**, **526** of the second mold half **112** remain at least partially inserted in the corresponding openings in the first mold half **110** throughout the injection molding process. By positioning the arm tool **114** in the first position **200** to minimize the distance between the mold halves **110**, **112** required to insert arm tool **114** between the mold halves **110**, **112**, the alignment pins **510**, **512**, **514**, **516**, **520**, **522**, **524**, **526** of the second mold half **112** can remain partially inserted in the openings of the first mold half **110** during de-molding. Leaving the alignment pins **510**, **512**, **514**, **516**, **520**, **522**, **524**, **526** of the second mold half **112** partially inserted in the first mold half **110** during de-molding reduces the risk of misalignment of the mold halves **110**, **112** and reduces the risk of damage to the mold halves **110**, **112**.

(37) The second mold half **112** also includes ejector pins (not shown) that are used to eject the formed dialyzer housings **424** from the second mold half **112**. As described in further detail herein, the controller **160** coordinates the movement of the ejector pins and the application of suction through the vacuum line **152** to the arm tool **114** such that the ejector pins eject the dialyzer housings **424** from the mold **112** and the arm tool **114** provides suction and couples to the dialyzer housings **424** simultaneously.

(38) A method of manufacturing and packing dialyzer housings will now be described with references to FIGS. 6-21.

(39) As depicted in FIG. 6, the mold halves **110**, **112** are closed while the injection molding device **102** performs injection molding of a pair of dialyzer housings (not shown). While the injection molding device **102** performs injection molding of the dialyzer housings, the robotic arm **104** is in a retracted position such that the arm tool **114** is positioned over the mold halves **110**, **112**. As depicted in FIG. 6, the arm tool **114** is in the first position **200** such that the posts **214**, **216** of the U-shaped platform and the second set of suction cups **204** are pointing downward, and the first set of suction cups **202** are oriented laterally relative to the vertical projection **120**.

(40) Referring to FIG. 7, once injection molding of the dialyzer housings by the injection molding device **102** is complete, the second mold half **112** retracts and moves apart from the first mold half **110** to expose the dialyzer housings. The distance between the mold halves **110**, **112** is sized to allow the arm tool **114** in the first position **200** to be inserted between the mold halves **110**, **112** with the first set of suction cups **202** facing the second mold half **112**. For example, the distance between the mold halves **110**, **112** in the open position following injection molding is between about 230 mm to accommodate the insertion of the arm tool **114** positioned in the first position **200** between the mold halves **110**, **112**. As previously discussed, the alignment pins **510**, **512**, **514**, **516**, **520**, **522**, **524**, **526** of the second mold half **112** remain partially inserted in the openings of the first mold half **110** through the entire manufacturing process. The controller **160** controls the movement of the second mold half **112** to move apart from the first mold half **110** by a predetermined distance to open the mold.

(41) As depicted in FIG. 8, once the injection molding process is complete and the controller determines, based on signals received from rotary encoders (not shown) in the injection molding device **102**, that the second mold half **112** has moved a predetermined amount (e.g., about 230 mm) to expose the dialyzer housings, the controller **160** for the injection molding device **102** transmits a signal to the controller **162** for the robotic arm **104** to indicate that the mold **110**, **112** is open. In response to receiving a signal from the controller **160** of the injection molding device **102** indicating that the mold **110**, **112** is open, the controller **162** of the robotic arm **104** controls the vertical projection **120** of the robotic arm **104** to extend to insert the arm tool **114** between the first mold half **110** and the second mold half **112**. The vertical projection **120** of the robotic arm **114** continues to extend until the controller **162** for the robotic arm **104** determines, based on feedback received from rotary encoders (not shown) of the robotic arm **104**, that the vertical projection **120** has extended a predetermined distance that corresponds to the first pair of suction cups **208** of the first set **202** being vertically aligned with a first dialyzer housing in the second mold half **112** and the second pair of suction cups **210** of the first set **202** being vertically aligned with a second dialyzer housing in the second mold half **112**.

(42) FIG. 9 depicts a side view of the molding device **102** with the arm tool **114** inserted between the mold halves **110**, **112** of the molding device **102**. As can be seen in FIG. 9, the first pair of suction cups **208** of the first set **202** is vertically aligned with a first dialyzer housing **902** in the second mold half **112**. In addition, the second pair of suction cups **210** (not shown) of the first set **202** is vertically aligned with a second dialyzer housing **904** in the second mold half **112**. As previously discussed, the controller **162** can determine the position of the arm tool **114** in three-dimensional space based on signals received from rotary encoders (not shown) of the robotic arm **104**.

(43) Once the controller has reached the fixed spatial location corresponding to alignment between the pairs of suction cups **208**, **210** and dialyzer housings **902**, **904**, the controller **162** ceases extension of the vertical projection **120**.

(44) Once the pairs of suction cups **208**, **210** are vertically aligned with the dialyzer housings **902**, **904**, the vertical projection **120** travels along the lateral boom **118** of the robotic arm **104** until each of the suction cups in the pairs **208**, **210** are moved into contact with the surface of the dialyzer

housings **902, 904**. For example, once the first set of suction cups **202** are vertically aligned with the dialyzer housings **902, 904**, as determined by the controller **162** based on rotary encoder signals, the controller **162** controls the vertical projection **120** of the robotic arm **114** to travel laterally along the lateral boom **118** towards the second mold half **112**. The vertical projection **120** continues to move towards the second mold half **112** until the controller **162** determines, based on signals received from the rotary encoder(s) of the robotic arm **104**, that the coordinates of the arm tool **114** correspond to a predetermined position corresponding to the first and second pairs of suction cups **208, 210** being in contact with the surface of the dialyzer housings **902, 904**, respectively.

(45) Once the pairs of suction cups **208, 210** are each aligned with and positioned proximate the dialyzer housings **902, 904**, the controller **162** initiates the application of suction through the pairs of suction cups **208, 210** and sends a signal to the controller **160** of the injection molding device **102** to move the ejector pins (not shown) of the second mold half **112**. The extension of the ejector pins outwards from the second mold half **112** forces the dialyzer housings **902, 904** out of the corresponding cavities in the second mold half **112**. As the ejector pins of the second mold half **112** force the dialyzer housings **902, 904** out of the mold half **112**, vacuum suction is applied through each of the pairs of suction cups **208, 210** to couple the dialyzer housings **902, 904** to the first and second pairs of suction cups **208, 210**, respectively. As previously discussed, each suction cup in the pairs **208, 210** includes an opening through its center, and each suction cup in the pairs **208, 210** is fluidly coupled to a vacuum source (e.g., vacuum source **150** of FIG. 1) via vacuum lines **152, 154, 156**, enabling suction to be applied through the center of each suction cup. The suction applied through the pairs of suction cups **208, 210** is transferred to the surface of the dialyzer housings **902, 904**, coupling the housings **902, 904** to the pairs of suction cups **208, 210** in the first set **202**.

(46) Referring to FIG. 10, once each of the dialyzer housings **902, 904** are removed from the second mold half **112** and coupled to the first set of suction cups **202** via vacuum suction, the vertical projection **120** of the robotic arm **104** is retracted and suction is continually supplied through the first set of suction cups **202** to lift the dialyzer housings **902, 904** out of the injection molding device **102**. In some examples, the controller **160** determines that the dialyzer housings **902, 904** are coupled to the first set of suction cups **202** based on a signal received from a vacuum confirmation sensor (not shown) on the arm tool **114**. Once the vertical projection **120** has retraced to a predetermined position to lift the dialyzer housings **902, 904** out of the injection molding device **102**, the controller **162** of the robotic arm **104** sends a signal to the controller **160** of injection molding device **102**. In response to receiving the signal from controller **162**, the controller **160** of the injection molding device **102** controls the second mold half **112** to move the second mold half **112** towards the first mold half **110** a predetermined distance corresponding to the halves **110, 112** touching and the mold being closed, as depicted in FIG. 10. In addition, as the vertical projection **120** retracts, suction is continually applied through the first set of suction cups **202** to maintain the coupling of the dialyzer housings **902, 904** to the arm tool **114**.

(47) After lifting the dialyzer housings **902, 904** out of the injection molding device **102**, the robotic arm **104** and arm tool **114** are used to place the dialyzer housings **902, 904** on a cooling table **106**. FIGS. 11-15 depict a process of placing the dialyzer housings **902, 904** on the cooling table **106** using the arm tool **114**.

(48) Referring to FIG. 11, with the dialyzer housings **902, 904** coupled to the first set of suction cups **202** of the arm tool **114** via vacuum suction, the lateral boom **118** of the robotic arm **104** travels forward a predetermined distance along the base **116** of the robotic arm **104**. As previously discussed, the base **116** includes a set of tracks **122** along its length to allow for smooth movement of the lateral boom **118** forward and backward along the base **116**. The lateral boom **118** continues to travel forward along the base **116** until the controller **160** receives a signal from the rotary encoders of the robotic arm **104** indicating that the lateral boom **118** has travelled a predetermined distance corresponding to the arm tool **114** being positioned such that the first set of suction cups **202** are positioned over a pair of empty cooling racks **918, 920** on the cooling table **106**.

(49) Still referring to FIG. 11, in addition to the lateral boom 118 travelling along the base 116, the vertical projection 120 travels laterally along the lateral boom 118 a predetermined distance to position the arm tool 114 such that the first set of suction cups 202 are positioned over a pair of empty cooling racks 918, 920 on the cooling table 106. For example, the vertical projection 120 continues to travel laterally along the lateral boom 118 until the controller 160 receives a signal from rotary encoders of the robotic arm 104 indicating that the vertical projection 120 has travelled a predetermined distance corresponding to the arm tool 114 being positioned such that the first set of suction cups 202 are positioned over a pair of empty cooling racks 918, 920 on the cooling table 106.

(50) As the robotic arm 104 moves to position the arm tool 114 over the cooling racks 918, 920, the arm tool 114 rotates about the pin connector 302 to rotate the arm tool 114 from the first position 200 (as depicted in FIG. 10) to the second position 300 (as depicted in FIG. 12). FIG. 11 depicts the rotation of the arm tool between the first position 200 and the second position 300 as the robotic arm 104 travels towards the cooling table 106. As previously discussed, the arm tool 114 includes a pneumatic cylinder (not shown) that applies a force to an end of the arm tool 114 and causes the arm tool to rotate about the pin connector 302. The controller 160 coordinates the rotation of the arm tool 114 between the first fixed position 200 and the second fixed position 300 based on the current location of the arm tool 114 in three-dimensional space, as determined based on the signals received from the rotary encoders of the arm tool 114. For example, the controller 162 is programmed to rotate the arm tool 114 from the first position 200 to the second position 300 at a specific point in the sequence of movements of the manufacturing cycle and based on the position of the arm tool 114 in three-dimensional space.

(51) FIG. 12 depicts the arm tool 114 positioned in the second position 300 with the dialyzer housings 902, 904 coupled to the first set of suction cups 202, and the first set of suction cups 202 facing downwards and aligned over the empty cooling racks 918, 920 on the cooling table 106.

(52) Referring to FIG. 13, once the arm tool 114 is in the second position 300 with the dialyzer housings 902, 904 aligned over the empty cooling racks 918, 920, as determined based on signals received by the controller 162 from the rotary encoder(s) of the robotic arm 104, the vertical projection 120 of the robotic arm 104 extends a predetermined amount (as detected by the rotary encoder(s) of the robotic arm) to lower the dialyzer housings 902, 904 into the respective cooling racks 918, 920. Once the robotic arm 104 has lowered the dialyzer housings 902, 904 a predetermined amount into the cooling racks 918, 920, the vacuum suction provided through the first set of suction cups 202 is stopped, which decouples the dialyzer housings 902, 904 from the arm tool 114 and releases the housings 902, 904 onto the cooling racks 918, 920. FIG. 14 depicts a perspective view of the arm tool 114 in the second position 300 releasing a pair of dialyzer housing 1002, 1004 into a pair of cooling racks 1018, 1020 on a cooling table 106.

(53) Once the dialyzer housings 902, 904 have been placed on the cooling table 106 and released from the arm tool 114, the arm tool 114 is used to move another pair of dialyzer housings into the storage container 108. FIGS. 15-21 depict a process of moving a pair of dialyzer housings from the cooling table 106 to the storage container 108.

(54) Referring to FIG. 15, once the dialyzer housings 902, 904 have been placed on the cooling racks 918, 920 and released from the arm tool 114, the vertical projection 120 of the robotic arm 104 retracts to lift the arm tool 114 above the cooling table 106. Once lifted above the cooling table 106 a predetermined distance, as measured by rotary encoders of the robotic arm 102, the controller 162 controls the arm tool 114 to rotate about the pin connector 302 from the second position 300 (as depicted in FIG. 15) to the first position 200 (as depicted in FIG. 16). As previously discussed, the arm tool 114 includes a pneumatic cylinder (not shown) that applies a force to an end of the arm tool 114 and causes the arm tool to rotate about the pin connector 302. The controller 160 coordinates the rotation of the arm tool 114 from the second fixed position 300 to the first fixed position 200 based on the spatial positioning of the arm tool 114. For example, the controller 162 is

programmed to rotate the arm tool **114** from the second position **300** to the first position **200** at a specific point in the sequence of movements of the manufacturing cycle and based on the position of the arm tool **114** in three-dimensional space.

(55) Referring to FIG. **17**, once the arm tool **114** is in the first position with the second set of suction cups **204** facing downwards towards the cooling table **106**, the vertical projection **120** of the robotic arm **104** translates along the lateral boom **118** until the controller **160** receives a signal with coordinates indicating that the second set of suction cups **204** is positioned over the center of a second pair of dialyzer housings **906**, **908** on the cooling table **106**. In addition, if necessary, the lateral boom **118** of the robotic arm **104** travels along the tracks **122** of the base **116** of the robotic arm **104** until the controller **160** receives a signal from the rotary encoder(s) that indicates that the robot has reached a position with corresponding with the second set of suction cups **204** being positioned over the center of the second pair of dialyzer housings **906**, **908** on the cooling table **106**. As depicted in FIG. **17**, the first pair of suction cups **220** of the second set **202** is aligned with the first dialyzer housing **906**, and the second pair of suction cups **222** of the second set **202** is aligned with the second dialyzer housing **908** to enable each pair of suction cups **220**, **222** to couple with the respective dialyzer housing **906**, **908**.

(56) Still referring to FIG. **17**, once the pairs of suction cups **220**, **222** of the second set **204** are positioned over the centers of the respective dialyzer housings **906**, **908**, the controller **162** controls the vertical projection **120** of the robotic arm **104** to extend a predetermined distance to lower the arm tool **114** towards the dialyzer housings **906**, **908**. In addition, the controller **162** controls vacuum suction to be applied through the vacuum lines **152**, **154**, **156** to the pairs of suction cups **220**, **222**, and through the center of each suction cup in the pairs of suction cups **220**, **222** in order to couple the dialyzer housings **906**, **908** to the pairs of suction cups **220**, **222**. The vertical projection **120** continues to extend until the controller **160** receives a signal from the rotary encoders indicating that the vertical projection **120** has extended a predetermined distance and receives a signal from a vacuum confirmation sensor indicating that the first and second pairs of suction cups **220**, **222** are in contact with the surface of the dialyzer housings **906**, **908** and coupled to the dialyzer housings **906**, **908**, respectively.

(57) As previously discussed, each suction cup in the pairs **220**, **222** includes an opening through its center, and each suction cup in the pairs **220**, **222** is fluidly coupled to a vacuum source (e.g., vacuum source **150** of FIG. **1**) via vacuum lines **152**, **154**, **156**, enabling suction to be applied through the center of each suction cup. The suction force applied through the pairs of suction cups **220**, **222** is transferred to the surface of the dialyzer housings **906**, **908**, which couples the dialyzer housings **906**, **908** to the suction cup pairs **220**, **222**. FIG. **18** depicts a perspective view of the arm tool **114** positioned to couple a pair of dialyzer housings **1006**, **1008** to the first and second pairs of suction cups **220**, **222**, respectively, of the second set of suction cups **204** in order to remove the dialyzer housings **1006**, **1008** from the cooling table **106**.

(58) Referring to FIG. **19**, once suction has been applied to the dialyzer housings **906**, **908** to couple the dialyzer housings **906**, **908** to the pairs of suction cups **220**, **222**, as determined based on a signal received by the controller **162** from a vacuum confirmation sensor (not shown), the controller **162** controls the vertical projection **120** of the robotic arm **104** to retract to lift the dialyzer housings **906**, **908** off the cooling table **106**. As the vertical projection **120** retracts, suction is continually applied through the pairs of suction cups **220**, **222** to maintain the coupling of the dialyzer housings **906**, **908** to the arm tool **114**.

(59) The vertical projection **120** travels along the lateral boom **118** of the robotic arm **104** until the controller **162** receives a signal from the rotary encoder(s) that the vertical projection **120** has travelled a predetermined distance corresponding with a position of the arm tool **114** in three-dimensional space that positions the second set of suction cups **204** over the storage container **108**, as depicted in FIG. **20**. In addition, if necessary, the lateral boom **118** of the robotic arm **104** travels along the tracks **122** of the base **116** of the robotic arm **104** until the controller **160** receives a signal

from the rotary encoder(s) that the lateral boom **118** has travelled a predetermined distance corresponding with a position of the arm tool **114** in three-dimensional space that positions the second set of suction cups **204** over the storage container **108**.

(60) Referring to FIG. **21**, once the arm tool **114** is positioned over the storage container **108**, the vertical projection **120** of the robotic arm **104** extends a predetermined amount, as determined by the rotary encoders of the robotic arm **102**, to lower the dialyzer housings **906**, **908** into the storage container **108**. The controller controls the movement of the robotic arm **104** components to position the housings **906** and **908** at has a programmed position in three-dimensional space. Once the housings **906**, **908** are positioned in the predetermined position, as determined based on the position of the arm tool **114** determined based on signals transmitted by the rotary encoders of the robotic arm **104**, the controller **162** stops the application of the vacuum suction through the second set of suction cups **204** to decouple the housings **906**, **908** from the suction cups **204** and place the housings **906**, **908** in the predetermined position in the container **108**. As the container **108** fills with dialyzer housings, the controller **162** tallies the number of housings placed in the container to determine the unique position for placing each housing in the spatial volume of the container **108**. By coupling the cooled dialyzer housings **906**, **908** to the second set of suction cups **204** with the arm tool **114** in the first position **200**, the length of the posts **214**, **216** is maximized, which allows the dialyzer housings **906**, **908** to be placed at the bottom of the storage container **108** without dropping the dialyzer housings **906**, **908** a significant distance. As such, the risk of damage to the dialyzer housings **906**, **908** during packing of the housings **906**, **908** in the storage container **108** is reduced.

(61) Before being moved into the storage container **108**, the surface of the second pair of dialyzer housings **906**, **908** is allowed to cool on the cooling table **106** to a temperature ranging between about 200° F. to about 250° F. In some examples, the dialyzer housings **906**, **908** each rest on the cooling table **106** about 51 seconds to about 70 seconds before being packed in the storage container **108**.

(62) Once the dialyzer housings **906**, **908** have been placed in the storage container **108** and released from the arm tool **114**, the vertical projection **120** of the robotic arm **104** retracts to raise the arm tool **114** out of the storage container **108**. In some implementations, after raising the arm tool **114** out of the storage container **108**, the vertical projection **120** moves along the lateral boom **118** and the lateral boom **118** moves along the base **116** to reposition the robotic arm **104** and arm tool **114** over the injection molding device **102** in preparation for retrieving another set of dialyzer housings from the injection molding device **102**. For example, after raising the arm tool **114** out of the storage container **108**, the robotic arm moves to the position depicted in FIG. **6** in preparation for retrieving another pair of dialyzer housings from the injection molding device **102**.

(63) This dialyzer housing manufacturing process continues until the storage container **108** is filled with dialyzer housings. Once filled, the storage container **108** is replaced with a new, empty storage container, and the process continues. The filled storage container **108** can be used to pack or store the dialyzer housings within the storage container **108**.

(64) While certain embodiments have been described above, other embodiments are possible.

(65) For example, while the method of demolding and packing the dialyzer housings has been described as relying signals received from rotary encoders to determine the coordinates of the arm tool **114** in three-dimensional space in order to coordinate the movements of the injection molding device **102**, the movements of the robotic arm **104**, the rotation of the arm tool **114**, and the application of vacuum suction, alternatively, the movements of the injection molding device **102**, the robotic arm **104**, the rotation of the arm tool **114**, and/or the application of vacuum suction through the suction cups **202**, **204** can be coordinated based on timing. For example, the robotic arm **104** moves between each of the positions of the manufacturing cycle at a predictable rate. Therefore, the times at which rotation of the arm tool **114** between the first position **200** and the second position **300** should occur can be determined. Further, the times at which vacuum suction

should be applied through each of the sets of suction cups **202**, **204** to couple the dialyzer housings to the appropriate set of suction cups **202**, **204** can be determined. Based on this determination, the controller **160** can be programmed to automatically move the robotic arm **104**, rotate the arm tool **114**, and apply vacuum suction through the suction cup sets **202**, **204** at predetermined times throughout the manufacturing cycle.

(66) While the system **100** has been described as including a robotic arm **104** with a lateral boom **118** and vertical projection **120**, other types of robotic components may be used to position the arm tool **114** and perform the method of demolding and packing the dialyzer housings.

(67) While the arm tool **114** has been described as including rectangular platform **206** and a U-shaped platform **212** with particular dimensions, platforms of other sizes and shapes can be used to support the suction cups of the arm tool **114**. In addition, while the widths **270**, **370** of the profile of the arm tool **114** in the first and second positions **200**, **300** have been described as being about 16 cm to about 17 cm (e.g., about 16.5 cm) and about 35 cm to about 36 cm (e.g., about 35.5 cm), respectively, the arm tool **114** can be configured to have different profile widths in each position **200**, **300**.

(68) While the arm tool **114** has been described as including 8 total suction cups, other numbers of suction cups are possible. For example, in some implementations, the arm tool includes four total suction cups, with two suction cups in the first set **202** and two suction cups in the second set **204**. In this arrangement, one suction cup is coupled to each post **214**, **216** of the U-shaped platform **212**, and one suction cup is attached to each end of the rectangular platform **206**. Further, in implementations in which the arm tool **114** includes a total of four suction cups, a single suction cup is used to couple the arm tool **114** to a single dialyzer housing. Alternatively, the arm tool can include a greater number of suction cups (e.g., 12 total suction cups, 16 total suction cups, etc.).

(69) Further, while the arm tool **114** has been described as having an equal number of suction cups in the first set **202** and the second set **204**, alternatively, the first set of suction cups **202** and the second set of suction cups **204** can each include a different number of suction cups.

(70) In addition, while the arm tool **114** has been described as having a first set of suction cups **202** that is oriented about 90 degrees relative to the second set of suction cups **204**, other orientations of the first and second set of suction cups can be used. For example, in some implementations, the first set of suction cups **202** is oriented about 70 degrees to about 110 degrees relative to the second set of suction cups **204**. In some implementations, the first set of suction cups **202** is oriented about 70 degrees to about 110 degrees relative to the second set of suction cups **204**.

(71) While the arm tool **114** has been described as being configured to couple to two of dialyzer housings simultaneously, alternatively, the arm tool **114** can be configured to couple to other numbers of dialyzer housings. In some implementations, the arm tool **114** can be configured to couple to a single dialyzer housing. For example, the arm tool **114** can include a total of two suction cups: a first suction cup coupled to a first portion of the frame of the tool (e.g., the rectangular platform **206**) and a second suction cup coupled to a second portion of the frame of the tool (e.g., the U-shaped platform **212**), and the tool **114** can be configured to couple to a single dialyzer housing. Alternatively, the arm tool **114** can be configured to couple to three or more dialyzer housings simultaneously.

(72) Similarly, while the injection molding device **102** has been described being configured to form two dialyzer housings simultaneously, alternatively, the injection molding device **102** may be configured to form a different number of dialyzer housings (e.g., 1, 3, 4, etc.).

(73) While the robotic arm tool **114** has been described as rotating about 90 degrees between a first position **200** and a second position **300**, the arm tool **114** can be controlled to rotate a different amount between the first position **200** and the second position **300**. For example, in some implementations, the arm tool **114** rotates between about 70 degrees and about 110 degrees between the first position **200** and the second position **300**.

(74) In addition, while the arm tool **114** has been described as rotating between a first fixed

position **200** and a second fixed position **300**, alternatively, the arm tool **114** can move fluidly between various positions during the manufacturing cycle without stopping at fixed positions.

(75) While the robotic arm **104** and arm tool **114** have been described as being used in a system for manufacturing dialyzer housings, alternatively, the robotic arm **104** and arm tool **114** can be used for manufacturing other items. For example, the robotic arm **104** and the arm tool **114** can be used to demold other types of components from an injection mold.

(76) While the arm tool **114** has been described as being coupled to the robotic arm **104** with a pin connector **302**, alternatively, other coupling mechanisms can be used to couple the arm tool **114** to the robotic arm **104**.

(77) While the injection molding device **102** has been described as having four mold alignment pins, alternatively, the injection molding device **102** may include a different number of mold alignment pins (e.g., 2, 3, 5, 6, etc.). Similarly, while the injection molding device **102** has been described as having four core alignment pins, alternatively, the injection molding device **102** may include a different number of core alignment pins (e.g., 2, 3, 5, 6, etc.).

(78) In addition, while the molding process has been described as moving the second mold half **112** about 230 mm apart from the first mold half **110**, the mold can open other distances. For example, in some implementations, the second mold half **112** is moved about 200 mm to about 240 mm apart from the first mold half **110** to accommodate the insertion of the arm tool **114** (positioned in the first position **200**) between the mold halves **110**, **112**.

(79) A number of embodiments have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the disclosure. Accordingly, other embodiments are within the scope of the following claims.

Claims

1. A dialyzer housing manufacturing system comprising: a molding device configured to mold a dialyzer housing; and a tool coupled to a robotic arm and configured to retrieve the dialyzer housing from the molding device after the dialyzer housing is molded, the tool comprising: a frame; a first suction cup connected to a first portion of the frame; and a second suction cup connected to a second portion of the frame, the second suction cup being oriented at an angle of about 70 degrees to about 110 degrees relative to the first suction cup such that the first and second suction cups face different directions, wherein the tool is rotatable about 70 degrees to about 110 degrees into a first position in which the second portion of the frame faces a first direction, wherein the tool is rotatable about 70 degrees to about 110 degrees into a second position in which the first portion of the frame faces the first direction, and wherein a width of the tool in the first position is less than a width of the tool in the second position.
2. The dialyzer housing manufacturing system of claim 1, wherein the first and second suction cups are fluidly coupled to a vacuum source.
3. The dialyzer housing manufacturing system of claim 2, wherein the tool further comprises a vacuum confirmation sensor configured to determine that the dialyzer housing is coupled to the first suction cup.
4. The dialyzer housing manufacturing system of claim 1, further comprising: a third suction cup connected to the first portion of the frame; a fourth suction cup connected to the first portion of the frame; a fifth suction cup connected to the first portion of the frame; a sixth suction cup connected to the second portion of the frame; a seventh suction cup connected to the second portion of the frame; and an eighth suction cup connected to the second portion of the frame, the sixth, seventh, and eighth suction cups being oriented about 70 degrees to about 110 degrees relative to the third, fourth, and fifth suction cups.
5. The dialyzer housing manufacturing system of claim 1, wherein the molding device is configured to mold two dialyzer housings.

6. The dialyzer housing manufacturing system of claim 5, wherein the tool is configured to simultaneously retrieve the two dialyzer housings from the molding device.
7. The dialyzer housing manufacturing system of claim 1, wherein the molding device is an injection molding device.
8. The dialyzer housing manufacturing system of claim 1, further comprising: a pneumatic cylinder; and a rotation pin, wherein: the rotation pin couples the tool to the robotic arm; and the tool is configured to rotate about the rotation pin in response to a force applied to the tool by the pneumatic cylinder.
9. The dialyzer housing manufacturing system of claim 1, wherein a width of the tool in the first position is about 16 cm to about 17 cm.
10. The dialyzer housing manufacturing system of claim 9, wherein the molding device is configured to open a pair of mold halves between about 200 mm to about 240 mm after molding the dialyzer housing.
11. The dialyzer housing manufacturing system of claim 1, wherein a width of the tool in the second position is about 35 cm to about 36 cm.
12. The dialyzer housing manufacturing system of claim 1, wherein: the molding device comprises an alignment pin coupled to a first half of a mold; and the alignment pin remains partially inserted into a second half of the mold when the mold is opened after molding the dialyzer housing.
13. The dialyzer housing manufacturing system of claim 1, further comprising a cooling table for cooling the dialyzer housing.
14. The dialyzer housing manufacturing system of claim 1, further comprising a storage container for storing the dialyzer housing.
15. The dialyzer housing manufacturing system of claim 1, wherein the tool is configured such that the first portion of the frame faces a second direction while the second portion of the frame faces the first direction.
16. The dialyzer housing manufacturing system of claim 1, wherein the second portion of the frame comprises a u-shaped platform.
17. A device for removing a dialyzer housing from a mold, the device comprising: a tool coupled to a robotic arm, the tool comprising: a frame; a first suction cup connected to a first portion of the frame; and a second suction cup connected to a second portion of the frame, the second suction cup being oriented at an angle of about 70 degrees to about 110 degrees relative to the first suction cup such that the first and second suction cups face different directions, wherein the tool is rotatable about 70 degrees to about 110 degrees into a first position in which the second portion of the frame faces a first direction, wherein the tool is rotatable about 70 degrees to about 110 degrees into a second position in which the first portion of the frame faces the first direction, and wherein a width of the tool in the first position is less than a width of the tool in the second position; and a pin rotatably coupling the tool to the robotic arm.
18. The device of claim 17, wherein the first and second suction cups are fluidly coupled to a vacuum source.
19. The device of claim 17, further comprising: a third suction cup connected to the first portion of the frame; a fourth suction cup connected to the first portion of the frame; a fifth suction cup connected to the first portion of the frame; a sixth suction cup connected to the second portion of the frame; a seventh suction cup connected to the second portion of the frame; and an eighth suction cup connected to the second portion of the frame, the sixth, seventh, and eighth suction cups being oriented about 70 degrees to about 110 degrees relative to the third, fourth, and fifth suction cups.
20. The device of claim 17, wherein the tool is configured to rotate about 70 degrees to about 110 degrees between the first position and the second position about the pin.
21. The device of claim 20, wherein a width of the tool in the first position is about 16 cm to about 17 cm.

22. The device of claim 20, wherein a width of the tool in the second position is about 35 cm to about 36 cm.
